Doppler Shift Distribution in Satellite Constellations

Akram Al-Hourani [1]

[1]Affiliation not available

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Akram Al-Hourani, *Senior Member, IEEE*

***Abstract*—Doppler shift has significant implications in Low Earth Orbit (LEO) constellations due to the rapid relative movement of satellites. This paper presents a novel approach based on stochastic geometry to capture the statistical distribution of Doppler shift in modern mega satellite constellations. We derive a simplified expression for both the Doppler shift s-curve and its maximum value, highlighting the dependency of Doppler shifts on the zenith angle of the closest approach during a satellite pass. Additionally, we present a stochastic model for the distribution of these zenith angles within the random constellation model. Furthermore, the statistical distribution of the Doppler bound is analytically formulated and validated through extensive Monte Carlo simulations of two major satellite constellations currently in orbit.**

***Index Terms*—Mega satellite constellations, radio Doppler shift, stochastic geometry, random satellite constellations.**

I. Introduction

MEGA satellite constellations are large networks of satellites working collectively to provide communication services from space [1]. The main objective of such constellations is to achieve global, or near global, wireless coverage and to augment terrestrial mobile services. Current examples of such massive Low Earth Orbit (LEO) constellations can be found in OneWeb, Starlink, and Iridium NEXT that are collectively having thousands of satellites, with many future constellation launches on the roadmap such as Project Kuiper. This vast number of satellites motivates the use of higher spectral bands, e.g., Ku and Ka band and even exploring Q/V bands [2]. Such high frequency bands and fast relative satellite-user motion result in extreme Doppler shifts. In addition, modern systems are increasingly relying on wideband OFDM waveform [3] to provide spectrally-efficient and broadband services, however OFDM is inherently sensitive to Doppler shifts. Interestingly, the resulting Doppler shifts themselves are currently being explored for joint communication and sensing applications [4], where the Doppler signature generated from multiple satellites can be exploited for localization applications [5]. Accordingly, understanding the expected range of Doppler shift provides insights for designers on better developing phase tracking loops, setting the signal acquisition bounds, and analyzing the overall adverse impacts on communication signal. Indeed, the characterization of Doppler shifts in Low Earth Orbit (LEO) satellites is not a new endeavor, as evidenced by previous studies [6]. However, a thorough examination of Doppler shift distribution within the framework of Mega constellations has not been attempted previously.

Large satellite constellations are typically designed according to a certain geometric layout. For example, Iridium and OneWeb constellations follow the Walker-Star configuration, while Starlink constellation follows Walker-Delta configuration but with multiple non-uniform orbits and other orbits in transition. Such density of satellites motivated the application of tools from stochastic geometry which deals with random point processes. This tool is well-matured and well-understood in terrestrial networks, but has only recently been applied to quantify the performance of satellite networks [7], [8] providing insights on many key performance indicators, such as the coverage probability, pass duration, and handover time. In order to facilitate the tractability of such analysis, the random constellation model has been proposed in [1] providing a tight approximation for the performance bounds. An example illustration of the random constellation is shown in Fig. 1 as seen from the perspective of a ground user where available satellites lie within what is termed the local satellite dome.

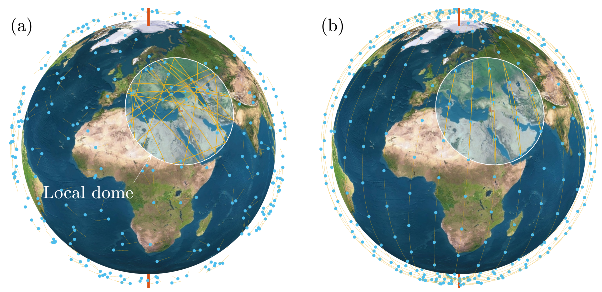

Fig. 1. Illustration of the random satellite constellation in (a) compared with traditional Walker-star constellation in (b).

Accordingly, given the significant Doppler effect in LEO orbits, this letter formulates the Doppler shift problem in modern Mega satellite constellations drawing tools from stochastic geometry using the random constellation model. Following are the main contributions of the paper: (i) it provides a concise formula describing the Doppler shift at any given phase of a LEO satellite pass, (ii) it formulates the bounds of the Doppler shift by only knowing the satellite altitude, (iii) it provides a closed-form expression of the statistical distribution of satellite's smallest zenith angle in an arbitrary pass, and finally (iv) it formulates a closed-form expression of Doppler shift bound distribution in a given constellation.

For convenience we list the main symbols and notations used in this paper in Table. I.

II. Geometric Model

Circular, or near-circular, orbits are the dominant types in LEO constellations providing consistent altitude path-loss towards the satellite sub-point. Accordingly, a satellite constellation can be thought to exist on an imaginary spherical shell with radius $a = h + R_\mathrm{e}$, where h is the altitude above mean sea level and R_e is the average radius of Earth. In such circular orbit, the angular speed of a satellite is steady and

A. Al-Hourani is with the School of Engineering, RMIT University, Melbourne 3000, VIC, Australia. Email: akram.hourani@rmit.edu.au

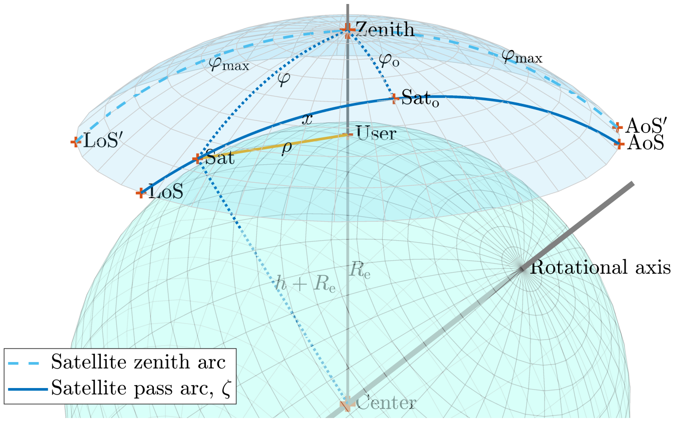

Fig. 2. Geometric model of a LEO satellite pass with respect to an arbitrary user, showing the point of acquisition of signal (AoS) and the Loss of Signal (LoS). The symbols are explained in Table I.

given by $\omega = \sqrt{\mu/a^3}$, where μ is the standard gravitational parameter of Earth.

Because of the Earth's curvature, a typical satellite cannot communicate with a user until it appears within the user's field-of-view (FoV). As such, only a spherical sector of the satellites' shell is relevant to the user terminal as illustrated in Fig. 2. This spherical sector, or FoV, has an opening angle of $2\varphi_{\max}$ as measured from the Earth center. Accordingly, a passing satellite will always have an Earth centered zenith angle $\varphi > \varphi_{\max}$ so it could establish a connection with the user terminal. Two factors determine $\varphi_{\max}$; (i) the Earth's curvature, and (ii) the antenna beamwidth ψ of the satellite, when it is the limiting factor rather than the Earth's curvature. Using geometric reasoning, the maximum zenith angle of a satellite can be found as follows,

$$\varphi_{\max} = \begin{cases} \operatorname{asin}\left(\frac{a}{R_{\mathrm{e}}}\sin\frac{\psi}{2}\right) - \frac{\psi}{2} & \psi \le \psi_{\mathrm{o}} \\ \operatorname{acos}\frac{R_{\mathrm{e}}}{a} & \psi > \psi_{\mathrm{o}} \end{cases}, \tag{1}$$

where $\psi_{\mathrm{o}} = 2\ \operatorname{asin}(R_{\mathrm{e}}/a)$ is the threshold beamwidth. A satellite enters the local dome of a given user at a point called the Acquisition-of-Signal-point (AoS) and exits at the Loss-of-Signal-point (LoS), also indicated in Fig. 2. Accordingly, the arc drawn by the satellite while it passes through the dome can be obtained by taking the spherical triangle determined by the three points **AoS**, **Zenith**, and **Sat**$_{\mathrm{o}}$ as indicated in the same figure. Thus, using the rule of cosines we can determine the angular measure of the pass arc as follows,

$$\zeta = 2\ \operatorname{acos}\left(\frac{\cos\varphi_{\max}}{\cos\varphi_{\mathrm{o}}}\right), \tag{2}$$

where φ_{o} is a key parameter that impacts the geometry of the pass arc; it represents the angular distance between the zenith and the satellite at its *closest approach* to the user, i.e. when the satellite at its closest angle to the zenith. For an arbitrary satellite pass we should always have $0 \le \varphi_{\mathrm{o}} \le \varphi_{\max}$ for the satellite to traverse within the FoV.

A. Distance between the satellite and the user terminal

Calculating the distance $\rho(t)$ between the satellite and the user terminal as function of time allows us to obtain the relative velocity. We start by taking the spherical triangle determined by the three points: **Sat**$_{\mathbf{o}}$, **Zenith**, and **Sat**, where **Sat** is the current position of the satellite at time t, and **Sat**$_{\mathbf{o}}$ is the position of the satellite at its closest approach to the user which is used as reference for time $t = 0$. These three points are shown in Fig. 2, thus by using the rule of spherical cosines we get,

$$\cos\varphi = \cos\varphi_{\mathrm{o}}\ \cos x, \tag{3}$$

where x is the section of the pass arc measured between Sat and Sat$_{\mathrm{o}}$, also note that the angle Zenith-Sat$_{\mathrm{o}}$-Sat is a right angle. Given the circular orbit assumption and ignoring the rotational motion of Earth during the short satellite pass, the angular measure of the pass arc section can be calculated based on the angular velocity ω as $x = \omega t$, where t is time measured from the closest approach point Sat$_{\mathrm{o}}$. Also by taking the triangle Sat-User-Earth Center as indicated in Fig. 2 the satellite-user distance can be obtained using the law of cosines as $\rho^2 = a^2 + R_{\mathrm{e}}^2 - 2aR_{\mathrm{e}}\cos\varphi$. Accordingly, by substituting from (3) we get,

$$\rho(t) \approx \sqrt{a^2 + R_{\mathrm{e}}^2 - 2aR_{\mathrm{e}}\cos\varphi_{\mathrm{o}}\ \cos\omega t}. \tag{4}$$

Note that this result ignores the rotation of Earth and thus only applies for fast pass durations, i.e., in LEO scenarios.

B. Doppler shift

Doppler shift is the change in the received radio frequency caused by the relative motion. For a narrow band carrier f_{o}, and a relative velocity v, the classical Doppler shift is given by, $f_{\mathrm{d}} = -\frac{v}{c}f_{\mathrm{o}}$, which is a good approximation for objects moving much slower than the speed of light c. In order to generalize the results we will use the normalized Doppler shift concept,

$$\nu = f_{\mathrm{d}}/f_{\mathrm{o}} = -v/c, \tag{5}$$

which is a unit-less quantity that can be measured in [Hz/Hz] or more conveniently in [Hz/MHz]. This means that the actual shift is $f_{\mathrm{d}} = f_{\mathrm{o}}\nu$. Since the velocity can be found by differentiating the distance, i.e. $v = \mathrm{d}\rho/\mathrm{d}t$, we deduce the normalized Doppler shift by differentiating (4) as follows,

$$\nu(t,\varphi_{\mathrm{o}}) = -\frac{a\omega R_{\mathrm{e}}}{c}\frac{\cos\varphi_{\mathrm{o}}\sin\omega t}{\sqrt{a^2 + R_{\mathrm{e}}^2 - 2aR_{\mathrm{e}}\cos\varphi_{\mathrm{o}}\ \cos\omega t}}. \tag{6}$$

In order to visualize the impact of the closest approach zenith angle φ_{o} on the Doppler shift we depict an example in Fig. 3 comparing (6) with simulation results that takes Earth's rotation and its oblateness into account.

The maximum Doppler occurs when the satellite is right at the edges of the pass arc, i.e., at the AoS and LoS. At these two points the satellite is at a position given by $\omega t = \pm\zeta/2$. By substituting from (1) and (2) into (6) we get the bounds of the normalized Doppler shift as follows,

$$\nu_{\max}(\varphi_{\mathrm{o}}) = \pm\frac{\omega R_{\mathrm{e}}}{c}\sqrt{\frac{a^2\cos^2\varphi_{\mathrm{o}} - R_{\mathrm{e}}^2}{a^2 - R_{\mathrm{e}}^2}}, \tag{7}$$

which depends on a given closest approach zenith angle φ_{o}. Accordingly, based on (7), the highest Doppler that could possibly occurs coincide when the satellite pass arc passes through

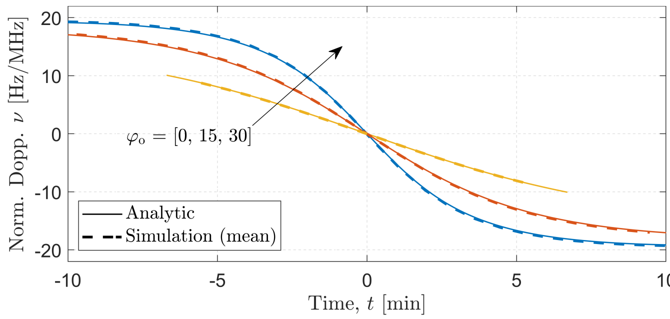

Fig. 3. Normalized Doppler shift as per the analytic formula in (6) comparing with simulation results (with a rotating and oblate Earth and using the random constellation model) at an altitude $h = 1,500$ km.

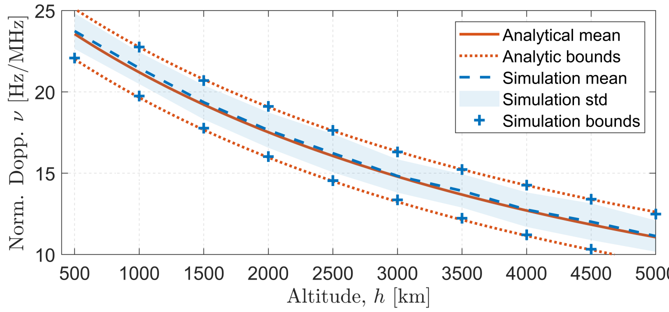

Fig. 4. The bounds of the absolute Doppler shift according to (8) compared with simulations that takes Earth's rotation into account. Satellite constellations: spherically wrapped binomial point process, users are randomly distributed around the Earth.

the zenith point, i.e., when $\varphi_o = 0$. Furthermore, a variation occurs to this maxima due to the rotation of Earth, the extreme effect of such rotation happens to users on the equator. Given the Earth linear speed of $v_e \approx 7.29 \times 10^{-5} \times R_e$, the resulting additional normalized Doppler is $v_e/c \approx 1.5497 \times 10^{-6}$. Accordingly, (4) reduces to a much simpler form,

$$\boxed{\nu_{\text{UB}} = \frac{R_e}{c}\sqrt{\frac{\mu}{(R_e+h)^3}} \pm 1.55 \quad \text{[Hz/MHz]}}\;, \tag{8}$$

representing the normalized bound of Doppler shift in LEO orbits, which is only affected by the selection of the satellite altitude. We depict in Fig. 4 the behavior of the normalized Doppler shift and compare it with actual simulations that take the Earth's rotation into account. Both the mean of the maxima and the bounds closely match with the practical simulation within a root mean square error (RMSE) $<$ 1.5 Hz/MHz.

C. Random constellation model

The random constellation is a theoretical model, that places satellites randomly on a spherical surface. Such model requires the satellites to maintain their average density while orbiting around Earth. One such random models is based on the spherical binomial process (also approximated into a Poisson point process) with homogeneous density of $\lambda = \frac{N_{\text{Sat}}}{4\pi a^2}$ [Satellites per unit area], where N_{Sat} is the total number of satellites in the shell.

Given the homogeneous nature of the process and without loss of generality, we can take the position of any user u as the typical location. For this user, any satellite that is currently passing within the local dome would have originated from

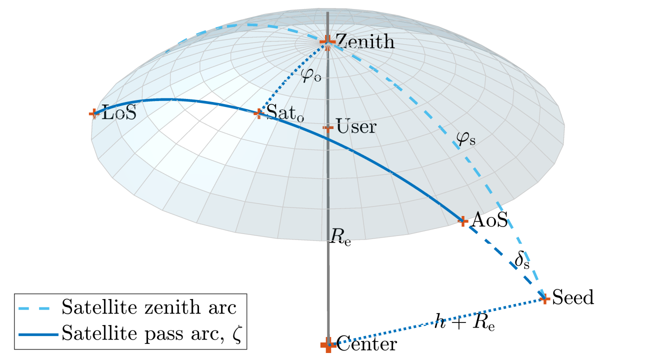

Fig. 5. Extended view of the local satellite dome and the concept of the seed point with respect to an arbitrary user.

some random zenith angle φ_s with an orbit angled δ_s with respect to the zenith arc, as indicated in Fig. 5, we call this location as the *seed*. Given the homogeneous density of the satellites, it can be shown that the distribution of the seed zenith angle needs to follow the below,

$$\begin{aligned} \varphi_s &\sim \frac{\pi}{2} - \operatorname{asin} \mathcal{U}(0,1) \\ \delta_s &\sim \mathcal{U}(0,\pi), \end{aligned} \tag{9}$$

where the uniform distribution of δ_s in the range $(0, \pi)$ is used given the symmetry of the orbit. As shown previously in Sec. II-B, the closest approach zenith angle φ_o is crucial in determining the Doppler profile of the satellite pass, accordingly, we derive the relation between φ_o and the seed angles by taking the spherical triangle determined by the three points: **Seed**, **Sat**$_o$, and **Zenith** as indicated in Fig. 5, and using the rule of sines we get,

$$\sin\varphi_o = \sin\delta_s \; \sin\varphi_s, \tag{10}$$

where the angle **Zenith**, **Sat**$_o$, **Seed** is $\pi/2$. Let's now derive the distribution of $Z = \sin\varphi_o$, based on (10) we have,

$$Z = XY, \;\; : \; Y = \sin\delta_s, \text{ and } X = \sin\varphi_s. \tag{11}$$

Given the homogeneous spherical binomial process in (9), and using the rule of variable transformation we can deduce the pdf of X and Y as follows,

$$\begin{aligned} f_X(x) &= \frac{x}{\sqrt{1-x^2}} & &: \; x \in (0,1), \\ f_Y(y) &= \frac{2}{\pi\sqrt{1-y^2}} & &: \; y \in (0,1). \end{aligned} \tag{12}$$

Given the independence of X and Y, the distribution of $Z = \sin\; \varphi_o$ results from the convolution of the two above pdfs obtained as follows,

$$\begin{aligned} f_Z(z) &= \int_z^1 f_X(x)\; f_Y\left(\frac{z}{x}\right) \frac{1}{|x|}\, dx, \\ &= \int_z^1 \frac{2}{\pi\sqrt{1-x^2}\sqrt{1-z^2/x^2}}\, dx = 1, \end{aligned} \tag{13}$$

which means a uniform distribution. Note that the lower limit of the integration is z because $X = Z/Y$ as per the relation in (10) and $0 \le Y \le 1$, accordingly $X \ge Z$.

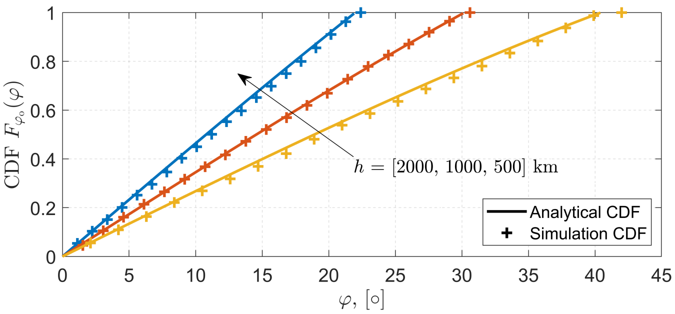

Fig. 6. Comparison between the distribution resulting from realistic simulation and the derived analytic formula in (16) at different constellation altitudes.

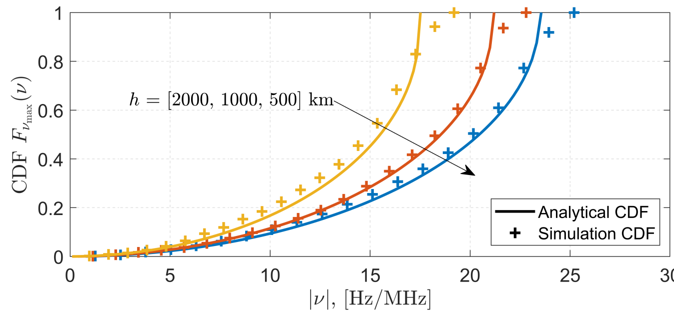

Fig. 7. The cumulative distribution function of the maximum Doppler shift during a pass of an arbitrary satellite in a given constellation with altitude h, comparing simulation results with the analytic formula in (20).

Using the variable transformation rule on (13) we can get the unconditional distribution of φ_o as follows,

$$f_{\varphi_o}(\varphi) = \cos\varphi \ : \ \varphi \in (0, \pi/2) \,. \tag{14}$$

However, considering the prior knowledge that the satellite has performed a pass through the local dome, we should restrict the above distribution such that the value of φ_o is bounded between 0 and $\varphi_{\max}$, i.e., $0 < \varphi_o \leq \varphi_{\max}$. This can be achieved by normalizing the pdf in (14) over the region of interest, accordingly the resulting conditional distribution is,

$$f_{\varphi_o}(\varphi) = \frac{\cos\varphi}{\sin\varphi_{\max}} \ : \ \varphi \in (0, \varphi_{\max}) \,, \tag{15}$$

where the denominator is the normalization factor obtained by integrating the unconditional pdf as follows $\int_0^{\varphi\max} \cos(\varphi) = \sin(\varphi_{\max})$. Accordingly, the CDF of the closest approach zenith angle is the integral of the pdf in (15), obtained as follows,

$$\boxed{F_{\varphi_o}(\varphi) = \frac{\sin\varphi}{\sin\varphi_{\max}} \ : \ \varphi \in (0, \varphi_{\max})} \,. \tag{16}$$

We depict the resulting cumulative distribution function (CDF) of φ_o in Fig. 6 and compare it with the simulation outcome at different constellation altitudes. In spite that the simulation accounts for realistic Earth rotation, we note a good conformity of the analytic solution, with a total RMSE $< 1.4\%$.

III. Doppler shift distribution

Given the random constellation model, we have derived the statistical distribution of the closest approach zenith angle φ_o in the previous section. However, our aim is to obtain the distribution of the maximum Doppler shift that could occur in an arbitrary satellite pass. To achieve this, we use the transformation of variables rule again. Let us now define $X = \cos\varphi_o$ where it can be shown based on (15) that the distribution of X is given by,

$$f_X(x) = \frac{x}{\varphi_{\max}\sqrt{1-x^2}} \ : \ x \in (\cos\varphi_{\max}, 1) \,. \tag{17}$$

Let's further define the Doppler bound function in (7) as $Y = g(X)$ where $Y = \nu_{\max}$. Given the rule of variables transformation is,

$$f_Y(y) = f_Y\left(g^{-1}(y)\right) \left| \frac{\mathrm{d}}{\mathrm{d}y} g^{-1}(y) \right| , \tag{18}$$

we start by inverting the function $g(.)$ as follows,

$$X = g^{-1}(Y) = \frac{1}{a}\sqrt{\frac{Y^2\left(a^2 - R_e^2\right)}{\left(\omega R_e/c\right)^2} + R_e^2}. \tag{19}$$

Thus, by applying the rule in (18) we get,

$$\boxed{f_{\nu_{\max}}(\nu) = \nu \frac{a^2 - R_e^2}{(\omega R_e/c)^2 a^2 \varphi_{\max}\sqrt{1 - \frac{(a^2-R_e^2)\nu^2}{(\omega R_e/c)^2 a^2} - \frac{Re^2}{a^2}}}} \,,$$
$$: \ \nu \in (0, \nu_{\text{UB}}) \ . \tag{20}$$

representing the PDF of the maximum Doppler shift during an arbitrary satellite pass which is only dependent on the constellation altitude embedded in $a = R_e + h$. Accordingly, the CDF representation can be obtained using numerical cumulative summation of (20). A comparison of the resulting CDF at different altitudes is shown in Fig. 7 along with the rotating Earth simulations with RMSE <2.5%.

IV. Simulation and Comparison

The simulation is performed in two main steps: (i) first is to create the geometry of the constellation and to propagate the satellite positions, (ii) second is to calculate the satellites-to-users distances and hence the resulting Doppler shifts caused by the movement.

Two constellation types were used in this analysis as shown in Fig. 1, namely the Walker-star and the random constellation. Walker-star is composed of N_p orbital planes each containing n satellites equally spaced along the orbit, these orbital planes are uniformly rotated around the z-axis of Earth with equal spacing. For the purpose of this simulation we have tested two practical constellations, namely (i) OneWeb (634 satellites), and (ii) Iridium and Iridium NEXT combined (111 satellites) directly using their updated two-line elements (TLE) table. The TLE table encodes the list of orbital elements of each satellite at a given epoch (time instance). Accordingly, the propagation of the satellite positions, i.e. moving the satellite across time, is performed by using the well-known Simplified General Perturbations Model (SGP4) which is a standard mathematical model for predicting the orbits of Earth's satellites based on their (TLEs).

The random constellation on the other hand is created first by randomly, and uniformly, scattering the satellites on a spherical shell with a desired altitude h, i.e. with radius a.

This can be achieved by drawing the right ascension (RA) and declination (DEC) angles from the following distributions,

$$\mathrm{RA_o} \sim \mathcal{U}[0, 2\pi], \quad \mathrm{DEC_o} = \mathrm{asin}\left(\sim \mathcal{U}[-1, 1]\right), \tag{21}$$

representing the initial points of the satellites. Secondly, each of these random initial points is assigned a random circular orbit with different roll angle δ_o. Such configuration preserves the homogeneous density of the satellites over time. In order to allow for the orbital propagation by common tools such as SGP-4 or simplified two-body propagators, we need to translate the initial angles $(\mathrm{RA_o}, \mathrm{DEC_o}, \delta_\mathrm{o})$ into the common six Keplerian elements. Using geometric reasoning, this relation can be found as follows,

$$i = \begin{cases} i_\mathrm{o} & \delta_\mathrm{o} \leq \pi \\ \pi - i_\mathrm{o} & \delta_\mathrm{o} > \pi \end{cases},$$

$$\Omega = \begin{cases} \mathrm{RA_o} - \Omega_o & \delta_\mathrm{o} \leq \pi,\ \mathrm{DEC_o} > 0 \\ \mathrm{RA_o} + \Omega_o - 2\pi & \delta_\mathrm{o} > \pi,\ \mathrm{DEC_o} > 0 \\ \mathrm{RA_o} + \Omega_o - 2\pi & \delta_\mathrm{o} \leq \pi,\ \mathrm{DEC_o} \leq 0 \\ \mathrm{RA_o} - \Omega_o & \delta_\mathrm{o} > \pi,\ \mathrm{DEC_o} \leq 0 \end{cases},$$

$$\omega = \begin{cases} 2\pi - \omega_\mathrm{o} & \mathrm{DEC_o} \leq 0 \\ \omega_\mathrm{o} & \mathrm{DEC_o} > 0 \end{cases},$$

where,

$$\begin{aligned} i_\mathrm{o} &= \mathrm{asin}\left(\sqrt{1 - \sin^2\delta_o\ \cos^2\mathrm{DEC_o}}\right), \\ \Omega_\mathrm{o} &= \mathrm{acos}\left(\frac{\cos\delta_\mathrm{o}}{\sqrt{1 - \sin^2\delta_o\ \cos^2\mathrm{DEC_o}}}\right), \\ \omega_\mathrm{o} &= \mathrm{acos}\left(\frac{\cos\mathrm{DEC_o}\cos\delta_\mathrm{o}}{\sqrt{1 - \sin^2\delta_o\ \cos^2\mathrm{DEC_o}}}\right), \end{aligned} \tag{22}$$

Ground user locations are similarly scattered around Earth, where their positions are also drawn randomly as follows,

$$\mathrm{LON_o} \sim \mathcal{U}[0, 2\pi], \quad \mathrm{LAT_o} = \mathrm{asin}\left(\sim \mathcal{U}[-1, 1]\right), \tag{23}$$

respectively representing the longitude and latitude of users.

After creating the geometry, we run the propagator as explained earlier and then collect distance measurements between satellites and underlying users. These measurements are filtered to accommodate only for the instances where satellites are visible to a given user for a full pass. The results of simulating the random constellation were demonstrated in Figs. 3, 4, 6, and 7 which were based on the random constellation model. Furthermore, a comparison with two real-life constellations; OneWeb and Iridium on one hand, and the analytic CDF of the Doppler bound on the other hand is shown in Fig. 8. The analytic and empirical curves demonstrate close resemblance with RMSE < 1.5 %.

V. Conclusion

This study offered an in-depth examination of Doppler shift effect in Low Earth Orbit (LEO) satellite constellations, elucidating the analytical bounds of Doppler shift that depend on the constellation altitude. By leveraging stochastic geometry for the spatial modeling of satellites in the random constellation model, we derived the statistical distribution of Doppler shift bounds. Comprehensive Monte Carlo simulations confirmed the analytical models, benchmarking them against prominent satellite constellations. This work significantly enhances understanding of Doppler effects in large-scale satellite networks, offering critical insights for the design, operation, and optimization of next-generation satellite communication systems.

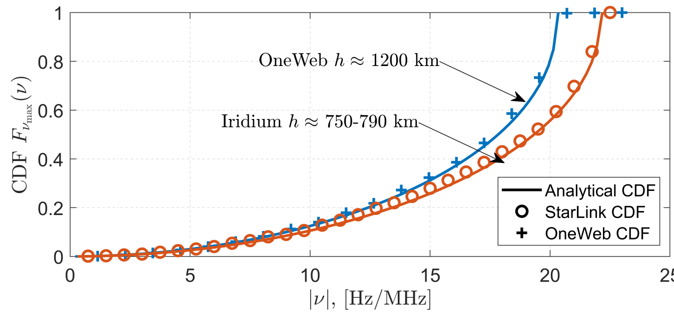

Fig. 8. Comparison between the analytic formula for the Doppler bound distribution provided in (20) with two real-life networks currently operational; OneWeb and Iridium.

TABLE I
Symbols and Parameters

Symbol	Parameter	Value
R_e	Earth average radius	6,371 km
μ	Earth standard gravitational parameter	$3.986{\times}10^{14}$ $\mathrm{m^3/s^2}$
h	Satellite altitude	150 - 2,000 km
a	Satellite orbit radius	$R_\mathrm{e} + h$
φ	Earth centered zenith angle	-
φ_o	φ at the closest approach	-
φ_max	φ at the rim of the FoV	-
ν	Normalized Doppler shift $f_\mathrm{d}/f_\mathrm{o}$	-
ζ	Pass arc from the closest approach	-
ρ	Satellite-user distance	-
f_o	Carrier frequency	-

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